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Sustainable seaweed aquaculture development in the Mekong Delta under climate change

Phu-Tho Nguyen[1,2*], Phuong-Loan Phan[1,2], Huu-Thanh Nguyen[1,2] and Huu-Tri Nguyen[1,2]

*Correspondence:
nptho@agu.edu.vn

[1] An Giang University, An Giang, Vietnam
[2] Vietnam National University, Ho Chi Minh City, Vietnam

Abstract

This structured narrative review synthesizes current knowledge on the development of seaweed aquaculture in the Mekong Delta under climate change. Literature searches were conducted through 30 June 2026 using Scopus, Web of Science, and Google Scholar, followed by screening and thematic synthesis. The Mekong Delta is increasingly exposed to sea-level rise, salinity intrusion, and extreme weather events, which are undermining the viability of conventional agricultural systems and rural livelihoods. In this context, identifying climate-resilient and sustainable production alternatives has become an urgent regional priority. Seaweed aquaculture has emerged as a promising nature-based option because of its potential to generate economic benefits, deliver ecosystem services, and enhance coastal resilience in low-lying and climate-vulnerable areas of the Delta. By integrating environmental, economic, and institutional perspectives, this review examines seaweed aquaculture as a climate-responsive and policy-relevant pathway for coastal development in the region. The review indicates that seaweed farming could support climate adaptation, nutrient removal, livelihood diversification, and broader ecosystem-service provision in the Mekong Delta. Current regional evidence most clearly documents the red seaweed *Gracilaria tenuistipitata* in improved-extensive shrimp and shrimp–crab pond systems. However, its expansion remains constrained by interconnected environmental, financial, institutional, and technological barriers. In response, the study proposes a policy-oriented framework for sustainable scaling, emphasizing that the long-term viability of seaweed aquaculture in the Mekong Delta depends on the effective alignment of technological innovation, market development, and institutional support. Overall, the review positions seaweed aquaculture not merely as an alternative production activity, but as a strategic option for sustainable coastal transformation under climate change.

Keywords: Seaweed aquaculture, Mekong Delta, Climate change adaptation, Ecosystem services, Sustainable development, Blue economy

1 Introduction

Located in southern Vietnam, the Mekong Delta is a low-lying, river-dominated region whose economy and coastal livelihoods are closely linked to land and water resources. For the purposes of this review, the coastal Mekong Delta comprises coastal areas within

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the provinces of Dong Thap, Vinh Long, Ca Mau, and An Giang, as well as Can Tho City, extending from the eastern estuarine zone along the East Sea to the Gulf of Thailand. The Mekong Delta is one of the most productive agricultural regions in Southeast Asia and plays a vital role in ensuring national food security while contributing significantly to global rice and aquaculture markets [1]. However, the region is increasingly vulnerable to the impacts of climate change, including sea-level rise, salinity intrusion, land subsidence, and more frequent extreme weather events, all of which have altered hydrological regimes and undermined ecosystem stability [2–4]. Anthropogenic land subsidence, driven largely by intensive groundwater extraction, further compounds these pressures, with annual subsidence rates reaching several centimetres in some locations and thereby increasing relative sea-level rise, inundation, and salinization risks across the low-lying Delta [5]. These changes are placing growing pressure on conventional agricultural systems, particularly in low-lying coastal areas where water scarcity and salinity intrusion have become more severe and less predictable [6].

Traditional production systems in the Mekong Delta, especially intensive rice cultivation and shrimp monoculture, are becoming increasingly constrained under changing climatic and hydrological conditions [7, 8]. Rice farming, which depends heavily on freshwater availability, is becoming less viable due to prolonged drought and saline water intrusion [6, 9]. During normal dry seasons, water with salinity above 4 g L^{-1} typically penetrates approximately 40–60 km inland, further increasing the vulnerability of freshwater-dependent rice production [10]. At the same time, shrimp monoculture, although economically attractive in the short term, has often been associated with environmental degradation, including mangrove loss, water pollution, and greater disease risks [11, 12]. Together, these pressures highlight the urgent need for alternative production models that are both environmentally sustainable and economically resilient [8, 13].

In this context, seaweed aquaculture has emerged as a promising nature-based solution that aligns with the goals of sustainable development and climate adaptation [14]. Seaweed farming provides multiple ecological and socioeconomic co-benefits, including carbon uptake, nutrient removal from eutrophic waters, and coastal protection through wave attenuation [15]. It also offers opportunities for income diversification and livelihood resilience among coastal communities, particularly in areas where the productivity of traditional agriculture and aquaculture is declining [16]. In addition, integrating seaweed into existing aquaculture systems, such as shrimp ponds, may improve resource efficiency and reduce environmental externalities [17].

Despite growing interest in seaweed aquaculture, existing studies on this sector in Vietnam and the broader region remain fragmented [18, 19]. Previous research has tended to examine ecological functions, market and value-chain issues, or governance and policy dimensions separately, rather than through an integrated lens [18, 19]. As a result, comprehensive analyses that connect these dimensions to assess the development potential of seaweed aquaculture in the Mekong Delta are still limited. This gap limits the evidence base needed to design coordinated strategies for the sustainable scaling of the sector under climate change [19]. Addressing this gap requires an integrated synthesis that connects environmental, economic, and institutional evidence within the specific context of the Mekong Delta. Accordingly, this review synthesizes current knowledge on seaweed aquaculture development in the Mekong Delta by integrating

these perspectives, with particular attention to its development potential, sustainability challenges, and policy-oriented solutions. In doing so, the study aims to provide a more comprehensive foundation for future research and policymaking on climate-resilient and sustainable coastal development in the region.

The remainder of this review is organized as follows. Section 2 describes the methodological approach, and Sect. 3 presents the conceptual background and integrated analytical framework. Sections 4., 5. and 6. examine the current status and development potential, key constraints, and strategic solutions, respectively. Section 7 discusses policy implications, limitations of the current evidence base, and future research priorities.

2 Methodological approach

This study adopted a structured narrative review approach to synthesize current knowledge on the development of seaweed aquaculture in the Mekong Delta under climate change. The review was not designed as a systematic review and was not preregistered. Accordingly, PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) 2020 items were considered only where applicable for reporting transparency, without implying full PRISMA compliance. The procedures for literature identification, screening, eligibility assessment, inclusion, and thematic synthesis were explicitly documented. The review covered the environmental, economic, technological, market, social, and institutional dimensions of seaweed aquaculture, with particular emphasis on evidence relevant to climate adaptation and sustainable coastal development in the Mekong Delta.

Initial literature searches were conducted in January 2025 and updated on 30 June 2026 using Scopus, Web of Science, and Google Scholar. Additional policy and technical documents were identified through searches of the official repositories and websites of the Food and Agriculture Organization of the United Nations, the World Bank, the World Wide Fund for Nature, the United Nations Conference on Trade and Development, and relevant Vietnamese institutions. The searches combined terms related to the production system, geographical scope, and analytical themes. In Scopus, the exact search syntax was: TITLE-ABS-KEY(("seaweed aquaculture" OR "seaweed farming" OR "seaweed cultivation") AND ("Mekong Delta" OR Vietnam) AND ("climate change" OR adaptation OR "ecosystem services" OR livelihood* OR "value chain" OR policy OR governance)). In Web of Science Core Collection, the corresponding syntax was: TS=(("seaweed aquaculture" OR "seaweed farming" OR "seaweed cultivation") AND ("Mekong Delta" OR Vietnam) AND ("climate change" OR adaptation OR "ecosystem services" OR livelihood* OR "value chain" OR policy OR governance)). In Google Scholar, where field tags and truncation functions are more limited, the adapted search string was: ("seaweed aquaculture" OR "seaweed farming" OR "seaweed cultivation") AND ("Mekong Delta" OR Vietnam) AND ("climate change" OR adaptation OR "ecosystem services" OR livelihood OR "value chain" OR policy OR governance). Transferable international evidence was additionally identified through thematic reference-list and citation searching without geographical restrictions.

No lower publication-year restriction was imposed because earlier foundational studies and policy documents were retained when they remained relevant to the conceptual or regional analysis. The initial database searches were restricted to publications

in English and Vietnamese. Eligible sources included peer-reviewed research articles, review articles, doctoral theses, and authoritative policy or technical reports. Publications were included when they provided substantive evidence on at least one of the following topics: the current status of seaweed aquaculture; economic and livelihood potential; ecosystem services and environmental effects; cultivation technologies; market and value-chain development; climate adaptation; or institutional and policy conditions. Duplicate records, publications unrelated to seaweed cultivation or sectoral development, non-substantive web materials, and documents without sufficient information for thematic analysis were excluded. Laboratory, biochemical, or taxonomic studies were considered only when their findings were directly relevant to cultivation suitability, value addition, product development, or sustainability assessment. Grey literature was not subjected to a formal risk-of-bias assessment tool; however, policy and technical reports were appraised prior to inclusion based on source authority, methodological transparency, geographical relevance, and sufficient substantive detail for thematic synthesis.

The literature search identified 358 records, including 322 records from Scopus, Web of Science, and Google Scholar and 36 additional records from institutional repositories, organizational websites, reference-list searching, and citation searching. Duplicate records were identified using the duplicate-detection function in EndNote X8 and subsequently verified manually by comparing titles, DOIs, author names, and publication years. After 79 duplicate records were removed, 279 records underwent title and abstract screening. Of these, 160 records were excluded because they were clearly unrelated to the geographical or thematic scope of the review. The full texts of 119 potentially relevant documents were subsequently assessed against the eligibility criteria. Sixty-two documents were excluded because they were outside the geographical or thematic scope ($n = 25$), had insufficient relevance to seaweed aquaculture development and sustainability ($n = 21$), contained insufficient substantive information for thematic synthesis ($n = 10$), or were unavailable in full text ($n = 6$). A final set of 57 sources was included in the narrative synthesis. The review process is summarized in Fig. 1.

For each included source, information was extracted on geographical coverage, publication type, study or document characteristics, cultivation system or seaweed species where applicable, principal findings, and relevance to the analytical framework. Evidence was interpreted according to its methodological transparency, source authority, geographical relevance, and consistency with the broader literature. During the thematic synthesis, greater interpretive weight was given to peer-reviewed and Mekong Delta-specific evidence, while Vietnam-level and international evidence was used cautiously to address regional knowledge gaps. The evidence was synthesized thematically across five areas: conceptual background; current development and potential; key challenges and constraints; strategic solutions; and policy and research implications. Because the included sources differed substantially in design, geographical setting, evidence type, and reported outcomes, quantitative meta-analysis was not appropriate.

As a structured narrative review, this study may not have captured every relevant publication, particularly locally published or non-indexed Vietnamese-language evidence. The inclusion of Google Scholar and institutional grey literature improved regional and policy coverage but introduced variation in source quality and reporting

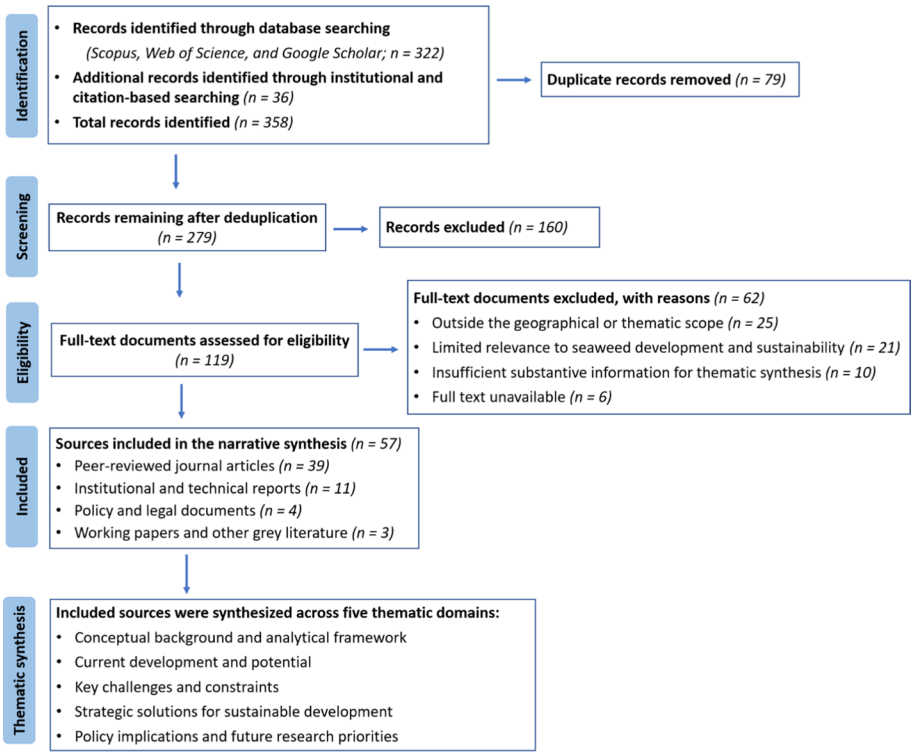


Fig. 1 Flow of literature identification, deduplication, screening, eligibility assessment, inclusion, and thematic synthesis for the structured narrative review. Source: Authors' compilation

detail. Moreover, evidence derived from locations outside the Mekong Delta may not be directly transferable to the region's distinctive ecological, production, market, and governance conditions. Accordingly, conclusions were framed cautiously, and quantitative or causal claims were avoided where they were not adequately supported by the available evidence.

3 Conceptual background: seaweed aquaculture and sustainability

3.1 Seaweed aquaculture as a nature-based solution

Seaweed aquaculture is increasingly recognized as a nature-based solution that can simultaneously address environmental degradation, climate change challenges, and livelihood sustainability [14]. In contrast to many conventional aquaculture systems that depend on external feed and other resource-intensive inputs, seaweed cultivation is a non-fed production system that requires little to no freshwater, fertilizer, or feed inputs, making it comparatively low-impact and resource-efficient [20].

One of the most important ecological functions of seaweed is its capacity to absorb carbon through photosynthesis and convert it into biomass, thereby contributing to climate change mitigation [15]. For quantitative context, Pessarrodona et al. [15] used a more realistic carbon-fixation value of approximately 350 g C m^{-2} y^{-1} based on the mean global harvested yield. Seaweed cultivation has also been discussed as a potential pathway for enhancing blue carbon benefits in coastal ecosystems; however, carbon fixation in cultivated biomass should not be equated directly

with long-term carbon sequestration, which remains highly context dependent [14, 21]. In addition, seaweed can remove excess nutrients, particularly nitrogen (N) and phosphorus (P), from aquatic environments, thereby helping to mitigate eutrophication and improve water quality [22]. This function is especially relevant in coastal areas affected by agricultural runoff and aquaculture effluents, where nutrient enrichment has become a persistent environmental concern [23].

Seaweed farming may also contribute to coastal protection by attenuating wave energy and supporting shoreline resilience [24, 25]. Model-based studies of suspended kelp canopies have evaluated shallow-water depths of approximately 2–4 m, with greater wave attenuation occurring in shallower water where the canopy occupies a larger proportion of the water column. Effective configurations generally involve dense longline canopies suspended near the surface, with longer and wider blades providing greater interaction with wave-driven flow; however, the optimal depth and canopy structure remain species- and site-specific [24, 25]. These ecosystem-based functions are particularly important in low-lying and climate-vulnerable coastal regions, where nature-based production systems can help strengthen both ecological stability and adaptive capacity. In this sense, seaweed aquaculture should not be understood merely as a source of biomass production, but as a multifunctional coastal activity with broader environmental value.

3.2 Contribution to sustainable development

Beyond its ecological functions, seaweed aquaculture also contributes to the broader goals of sustainable development across economic, social, and environmental dimensions [14, 18]. Economically, seaweed farming offers opportunities for income diversification, particularly for small-scale coastal households that are increasingly exposed to climate-related disruptions affecting conventional agriculture and aquaculture [16]. Its relatively low capital requirements and short production cycles may make it a more accessible livelihood option than many other forms of aquaculture [16]. In addition, rising demand for seaweed-based products, including food ingredients, pharmaceuticals, and bio-based materials, creates scope for value addition and market expansion [26].

From a social perspective, seaweed aquaculture may strengthen livelihood resilience by reducing dependence on a narrow set of climate-sensitive activities [27]. It can also support gender inclusion, as women often participate actively in cultivation, post-harvest handling, processing, and trading across seaweed value chains [28]. Such participation may generate wider community benefits and contribute to local adaptive capacity, especially in vulnerable coastal settings [14].

Environmentally, seaweed farming provides multiple ecosystem services, including water purification, habitat support for marine organisms, and carbon regulation [18]. These functions can improve ecological health while also supporting the sustainability of adjacent aquaculture and fisheries systems. Taken together, these contributions suggest that seaweed aquaculture has the potential to function not only as an economic activity, but also as a strategic pathway toward more resilient and sustainable coastal development.

3.3 Integrated analytical framework

To examine the role of seaweed aquaculture in sustainable development more systematically, an integrated analytical framework is needed to connect climate pressures, production responses, and sustainability outcomes [14, 29]. In this review, seaweed aquaculture is conceptualized as a pathway through which climate change pressures shape sector development and, in turn, influence economic, social, and environmental sustainability outcomes (Fig. 2).

Within this framework, climate change is understood not only as a source of vulnerability, but also as a driver of adaptation and innovation in coastal production systems [30]. Seaweed aquaculture is therefore framed as a potential adaptive response through which environmental constraints may be transformed into opportunities for more sustainable forms of production and livelihood development. Its outcomes may be reflected in improved income diversification, stronger social resilience, and enhanced environmental performance [14].

At the same time, the development of seaweed aquaculture does not depend on climate and ecological conditions alone. Its expansion is also shaped by technological capacity, market opportunities, and institutional support. Institutional support can moderate these sustainability outcomes through different pathways. Economically, clear regulations, extension services, market coordination, and access to finance can reduce investment uncertainty and improve producers' ability to capture value, whereas environmentally, effective zoning, monitoring, and production standards can help control ecological risks and strengthen the delivery of ecosystem services [29]. For this reason, the framework adopted in this study provides a foundation for analysing seaweed aquaculture not simply as a biological or technical activity, but as a socio-ecological development pathway. The analytical relationships discussed above are summarized in Fig. 2, which illustrates how climate change pressures shape seaweed aquaculture development and how sustainability outcomes are mediated by technological, market, and institutional conditions. Building on this analytical framework, the following section examines the current development status and the economic and environmental potential of seaweed aquaculture in the Mekong Delta.

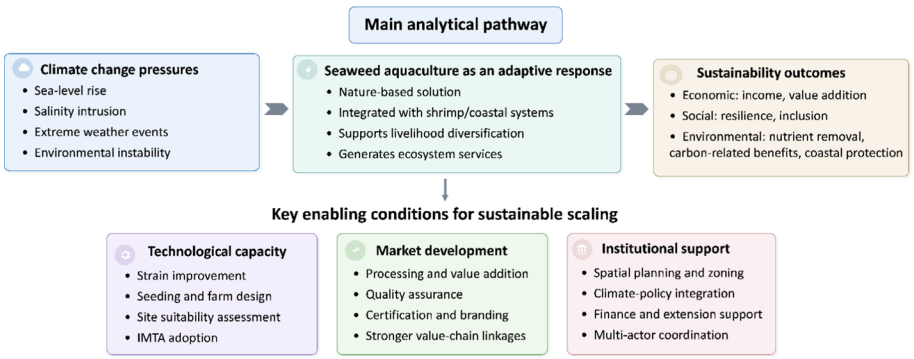


Fig. 2 Integrated analytical framework for seaweed aquaculture development in the Mekong Delta under climate change. Source: Authors' synthesis based on the reviewed literature

4 Current development and potential in the Mekong Delta

4.1 Current status

Compared with dominant aquaculture activities such as shrimp and fish farming, seaweed aquaculture in the Mekong Delta remains at an early stage of development. Although Vietnam cultivates several commercially important seaweed species, including *Kappaphycus alvarezii*, *Kappaphycus striatus*, *Eucheuma denticulatum*, and several *Gracilaria* taxa, large-scale production is still concentrated mainly outside the Delta [19, 31]. Within the Mekong Delta, current evidence most clearly documents the cultivation and management of the red seaweed *Gracilaria tenuistipitata*, particularly in improved-extensive shrimp and shrimp–crab pond systems [32, 33]. Other locally occurring macroalgae, including Cladophoraceae and *Enteromorpha* spp., are also found in shrimp ponds but are generally naturally occurring rather than established commercial crops [33]. Accordingly, *Kappaphycus* and *Eucheuma* species are better regarded as candidate taxa for future regional expansion than as major cultivated crops currently established in the Delta.

Where seaweed cultivation has been adopted, it is commonly integrated into shrimp-based farming systems, where it may improve resource efficiency and provide additional ecological functions [32]. However, these practices remain limited in scale and are not yet supported by sufficiently strong technical standardization, extension services, or development programs [19, 33]. Available evidence from the Mekong Delta primarily documents naturally occurring *Gracilaria tenuistipitata* biomass in improved-extensive shrimp ponds [33]. Vegetative propagation through retained thallus material is technically feasible for this species [34], but the reviewed literature did not identify dedicated commercial seaweed nurseries or hatcheries operating at regional scale in the Delta. As a result, seaweed aquaculture has so far made only a modest contribution to the wider aquaculture economy of the Mekong Delta, despite its potentially strategic relevance in the context of climate change and coastal livelihood transition.

4.2 Economic potential

Despite its currently limited scale, seaweed aquaculture has considerable economic potential in the Mekong Delta, particularly through value addition and closer integration with expanding global seaweed markets [26]. Seaweed biomass can be processed into high-value hydrocolloids such as agar and carrageenan, which are widely used in food, pharmaceutical, and industrial applications [35]. Seaweeds are also increasingly recognized as sources of bioactive compounds for functional foods, nutraceuticals, and pharmaceutical products [36]. In Vietnam, seaweed products currently serve domestic markets for human food and animal feed and are also used as raw materials for cosmetics and traditional medicine, while a proportion of production is directed toward international markets [19]. At the same time, growing demand for sustainable and bio-based materials is creating new market opportunities for seaweed-producing regions [26].

In the Mekong Delta, these trends suggest that the economic value of seaweed aquaculture lies not only in primary production, but also in its downstream processing and market diversification potential. Stronger investment in processing technologies, quality assurance systems, and certification could improve competitiveness and

facilitate entry into higher-value market segments [26]. At the local level, seaweed farming may also contribute to employment and income generation. Small-scale nearshore or pond-based cultivation may have relatively modest entry costs compared with many fed aquaculture systems [16]. However, transition toward commercially scaled production requires substantial investment in farm infrastructure, quality seedlings, post-harvest handling, processing facilities, certification, and market development [26]. In this regard, seaweed aquaculture may serve both as a livelihood diversification strategy and as an emerging component of a broader coastal bioeconomy.

4.3 Environmental potential

In addition to its economic prospects, seaweed aquaculture offers significant environmental potential that is highly relevant to the ecological challenges of the Mekong Delta [18]. One of its most important functions is the removal of excess nutrients, particularly N and P, from aquaculture effluents and agricultural runoff, thereby improving water quality and supporting the ecological sustainability of surrounding aquatic systems [22]. In laboratory experiments, nutrient uptake rates reported for *Gracilaria tenuistipitata* var. *liui* reached approximately 0.32–0.36 μmol N g^{-1} h^{-1} and 0.017–0.018 μmol P g^{-1} h^{-1} under favorable light conditions. Nutrient uptake varied with environmental factors including light intensity, pH, temperature, and salinity, with higher N and P uptake observed at temperatures of 21–26 °C [37]. By assimilating dissolved nutrients, seaweed cultivation may help reduce eutrophication in intensive farming areas and lower the risk of harmful algal blooms and related ecosystem degradation [23].

Seaweed aquaculture may also contribute to both climate change mitigation and adaptation through carbon uptake and climate-regulating functions [15]. In addition, its physical structure can attenuate wave energy and enhance coastal resilience, which is particularly valuable in low-lying coastal areas exposed to sea-level rise, erosion, and extreme weather events [24]. For the Mekong Delta, where environmental degradation and climate vulnerability are increasingly intertwined, these ecosystem-based functions strengthen the case for considering seaweed aquaculture not only as a productive activity, but also as part of a broader strategy for sustainable coastal management. Overall, the region possesses substantial yet underutilized potential for seaweed aquaculture, and realizing this potential will depend on the effective alignment of technological innovation, market development, and policy support.

5 Key challenges and constraints

Despite its considerable promise, the development of seaweed aquaculture in the Mekong Delta is constrained by a set of interrelated environmental, economic, institutional, and technological barriers. These constraints not only limit the scalability of the sector, but also undermine its long-term sustainability, productivity, and competitiveness. Understanding these challenges is therefore essential for assessing whether seaweed aquaculture can realistically evolve into a viable climate-resilient development pathway in the region.

5.1 Environmental constraints

Environmental pressures represent one of the most significant constraints on the expansion of seaweed aquaculture in the Mekong Delta. A major concern is the degradation of mangrove ecosystems associated with aquaculture expansion, particularly shrimp farming [13]. The conversion of mangrove forests into aquaculture ponds has contributed to biodiversity loss, weakened coastal protection, and increased greenhouse gas emissions, highlighting the trade-off between short-term economic gains and long-term ecological sustainability [38]. In addition, water pollution and chemical runoff from intensive agriculture and aquaculture pose serious risks to seaweed cultivation [39]. The discharge of nutrients, antibiotics, and pesticides into coastal waters can degrade water quality and disrupt surrounding ecosystems, especially in densely farmed areas where water exchange is limited and environmental monitoring remains weak [39]. These conditions may directly affect seaweed growth and overall farm performance. Moreover, because environmental contaminants, including pesticides and pharmaceuticals, can attach to or be taken up by seaweeds, agrochemical and antibiotic residues in exchanged pond water could compromise tissue quality and food safety [40].

The Mekong Delta is also highly exposed to climate-related stressors, including salinity fluctuations, temperature variability, and extreme weather events [41]. For *Gracilaria tenuistipitata*, the best-documented seaweed in local shrimp-based systems [32, 33], however, a locally validated upper thermal tolerance threshold has not yet been established. Consequently, the available regional evidence is insufficient to determine whether seasonal temperature peaks consistently exceed its physiological tolerance, although episodic heat stress may still reduce growth and cultivation stability. Such factors can alter seaweed physiology, reduce biomass yield, affect biochemical composition, and increase susceptibility to disease. As a result, environmental instability introduces significant uncertainty into cultivation systems and weakens production reliability, particularly for small-scale producers with limited adaptive capacity.

5.2 Economic and financial barriers

From an economic perspective, the development of seaweed aquaculture in the Mekong Delta is constrained by several structural barriers. One of the most important is limited access to capital and credit among small-scale producers, which restricts investment in cultivation infrastructure, quality seedlings, and post-harvest processing [16]. Formal collateral requirements can further restrict smallholders' access to credit, thereby limiting their capacity to invest in separate inlet and outlet canals, pumps, reservoirs, and filtration structures needed to improve pond-level water exchange and biosecurity [42]. High initial costs for farm establishment, equipment, and processing facilities may further discourage new entrants and limit the expansion of existing operations [26].

These financial constraints are compounded by market uncertainty, including price fluctuations, unstable demand in some commodity segments, and strong competition from established seaweed-producing countries [26]. Another major challenge is the weakness and fragmentation of seaweed value chains, as poor coordination among producers, processors, traders, and buyers reduces value addition and weakens farmers' bargaining power [26]. In many cases, producers remain dependent on intermediaries, which can compress profit margins and make income streams less stable and

predictable. Taken together, these barriers limit the economic attractiveness of seaweed aquaculture, especially in the absence of stronger downstream market support.

5.3 Institutional and policy gaps

Institutional and policy-related barriers also significantly constrain the development of the seaweed sector in the Mekong Delta [19]. One key issue is the lack of clear spatial planning and coherent governance arrangements for coastal aquaculture activities. At the local level, responsibilities for coastal aquaculture zoning and sea-area management are shared among provincial People's Committees, provincial Departments of Agriculture and Environment through their fisheries and marine–environmental management functions, and commune-level People's Committees for certain sea-area allocation and management responsibilities. This distribution of responsibilities can create coordination gaps in spatial planning, licensing, environmental management, and resource allocation, thereby discouraging investment and hindering long-term sector development [43]. In addition, technical standardization and operational guidance for safe production, quality assurance, and risk management remain insufficient in emerging seaweed value chains [43]. This limits producers' ability to meet market requirements, maintain consistent quality, and secure stable off-take agreements. Policy support for seaweed aquaculture also remains relatively limited compared with more established aquaculture sectors, reducing its attractiveness to both smallholders and private investors [19]. More broadly, seaweed aquaculture is still only partially integrated into marine spatial planning, climate adaptation policy, and coastal management strategies. This weak policy integration reduces the sector's visibility in public planning and limits access to the institutional support needed for sustainable expansion. For the Mekong Delta, where climate adaptation and coastal governance are increasingly interconnected, such policy gaps represent a major constraint on scaling seaweed aquaculture in a coordinated and strategic manner.

5.4 Knowledge and technology gaps

A further barrier to the sustainable expansion of seaweed aquaculture lies in the limited availability of scientific knowledge and technological innovation tailored to local conditions [19]. Existing research on seaweed cultivation in the Mekong Delta remains limited, particularly with regard to optimal farming techniques, species selection, and carrying-capacity assessment [31, 43]. This restricts the evidence base needed to support appropriate farm design and site-specific management decisions.

There is also still insufficient understanding of the ecological implications and long-term sustainability of large-scale seaweed farming, especially in relation to surrounding ecosystems and local environmental performance [44]. In addition, the potential of integrated systems such as integrated multi-trophic aquaculture (IMTA) remains underexplored in the regional context, despite their documented benefits in other settings [31, 45]. Equally important is the limited transfer of technology and practical knowledge to producers. Weak extension systems and insufficient training support can slow the adoption of improved farming practices, reduce productivity gains, and widen the gap between research and practice [46]. As a result, technological constraints in the Mekong Delta are not simply a matter of innovation availability, but also of adaptation,

dissemination, and local implementation capacity. Overall, addressing these knowledge and technology gaps will require stronger connections among research institutions, extension services, producers, and policymakers.

6 Strategic solutions for sustainable development

Unlocking the full potential of seaweed aquaculture in the Mekong Delta will require a coordinated and multidimensional strategy. Because the barriers facing the sector are environmental, economic, institutional, and technological in nature, the corresponding solutions must also be integrated across these domains. Sustainable development of the sector therefore depends not on a single intervention, but on the alignment of innovation, market development, governance support, capacity building, and climate adaptation.

6.1 Technological innovations

Technological innovation is a key prerequisite for improving productivity, resilience, and environmental performance in seaweed aquaculture [47]. One important priority is the development of disease- and climate-tolerant strains that can maintain yield stability under salinity fluctuations, temperature stress, and pathogen pressure [41]. Advances in selection, priming, and breeding technologies offer significant potential in this regard [41]. At the farm level, the adoption of improved cultivation practices is equally important. Optimized seeding methods, site-specific farm design, and better harvest timing can increase production efficiency while reducing environmental risks [47]. A particularly promising pathway is the expansion of IMTA, in which seaweed is co-cultivated with species such as shrimp or fish [17]. In such systems, seaweed functions as a natural biofilter by absorbing excess nutrients and improving water quality, thereby enhancing both production efficiency and ecological sustainability [17]. When IMTA is implemented with shrimp, biosecurity should include the use of health-screened stocking material, quarantine of new stocks where appropriate, filtration or screening of incoming water, routine health monitoring, and cleaning and disinfection of shared equipment to minimize pathogen introduction and cross-transmission among cultured components [45]. For the Mekong Delta, where shrimp-based systems are already widespread, the promotion of IMTA may provide a practical route toward more circular and resource-efficient aquaculture models.

6.2 Value chain development

Strengthening the seaweed value chain is essential for improving economic returns and market competitiveness [26]. A central priority is to improve linkages among farmers, processors, traders, and exporters, as better coordination can reduce inefficiencies, improve market transparency, and increase the bargaining power of primary producers [26]. Without stronger vertical and horizontal connections across the value chain, producers are likely to remain locked into low-value and unstable market positions.

Promoting value-added products is another critical strategy. Rather than relying primarily on raw or minimally processed seaweed, investment should support processing technologies that enable higher-value outputs such as hydrocolloids, functional foods, pharmaceuticals, and bio-based materials [26, 35]. This shift could

improve profit margins, diversify market outlets, and reduce dependence on lower-value commodity segments. In addition, branding and certification systems, including eco-labels and sustainability standards, can strengthen product credibility, facilitate market access, and improve compliance with quality and traceability requirements [48]. In the Mekong Delta, such measures would be particularly important for moving the sector from small-scale supplementary production toward a more competitive and commercially viable development model.

6.3 Policy and governance improvements

Effective policy and governance frameworks are essential for creating an enabling environment for sustainable seaweed development [49]. One of the most immediate priorities is the establishment of clear spatial planning and aquaculture zoning, which can reduce resource-use conflicts, improve spatial efficiency, and encourage long-term investment [43]. For a coastal region such as the Mekong Delta, where multiple production systems and environmental pressures intersect, spatial clarity is especially important.

Governments should also introduce targeted financial incentives and support mechanisms, including concessional credit, grants for technology adoption, and tailored assistance for smallholder producers, in order to lower entry barriers and stimulate investment across the sector [50]. At the same time, seaweed aquaculture should be more explicitly integrated into national and regional climate adaptation strategies, given its potential role as a nature-based solution within coastal development policy [50]. Better integration with broader marine spatial planning and coastal management frameworks would improve policy coherence, expand access to climate and development finance, and support more strategic sectoral scaling. In this sense, governance reform is not simply a matter of regulation, but of creating an institutional environment in which seaweed aquaculture can develop with clearer direction and stronger public support.

6.4 Capacity building and knowledge transfer

Human capacity development is a critical component of sustainable sector growth [14]. Training programs for farmers can strengthen technical skills in cultivation, harvesting, post-harvest handling, and farm management, thereby supporting higher productivity and more consistent product quality [50]. In emerging sectors such as seaweed aquaculture in the Mekong Delta, such capacity building is particularly important because many producers may have limited prior experience with specialized farming and processing techniques. Participatory research approaches that involve farmers, researchers, and local stakeholders can also facilitate the co-creation of knowledge and help adapt technological innovations to local conditions [51]. In parallel, stronger extension systems are needed to bridge the gap between research and practice. Effective extension can disseminate best practices, provide technical guidance, and encourage the adoption of improved technologies at scale [52]. Taken together, these measures can help ensure that innovation is not only developed, but also effectively translated into field-level practice.

6.5 Climate adaptation strategies

Given the high climate vulnerability of the Mekong Delta, adaptive strategies are essential to the long-term sustainability of seaweed aquaculture [30]. One key measure is the careful selection of cultivation sites based on environmental factors such as salinity, temperature, water depth, and nutrient availability, as site suitability strongly influences both productivity and risk exposure [53]. In this respect, site-selection planning should be treated as a foundational step rather than a purely technical afterthought.

The adoption of seasonal and adaptive farming practices is also important. Adjustments in cultivation timing, species choice, and farm management can improve resilience to climate variability and reduce production losses under changing environmental conditions. In addition, ecosystem-based management approaches can help integrate seaweed aquaculture into broader coastal systems by linking cultivation with mangrove conservation, water-quality protection, and the sustainable use of marine resources [54]. Such approaches are particularly relevant in the Mekong Delta, where the long-term viability of aquaculture depends on the health and resilience of interconnected coastal ecosystems. Overall, the sustainable development of seaweed aquaculture in the Mekong Delta requires a holistic strategy that combines technological innovation, value chain development, policy reform, capacity building, and climate adaptation. Only through the integration of these elements can seaweed farming evolve into a resilient, competitive, and environmentally sustainable sector. The current development and potential, key challenges and constraints, and corresponding strategic solutions discussed in Sects. 4., 5. and 6. are synthesized in Fig. 3. The organization of the figure follows the subsection structure of these three sections to facilitate cross-referencing.

Table 1 further synthesizes the main development opportunities, key constraints, strategic responses, and policy implications for seaweed aquaculture in the Mekong Delta. Building on this synthesis, the following section discusses the broader policy

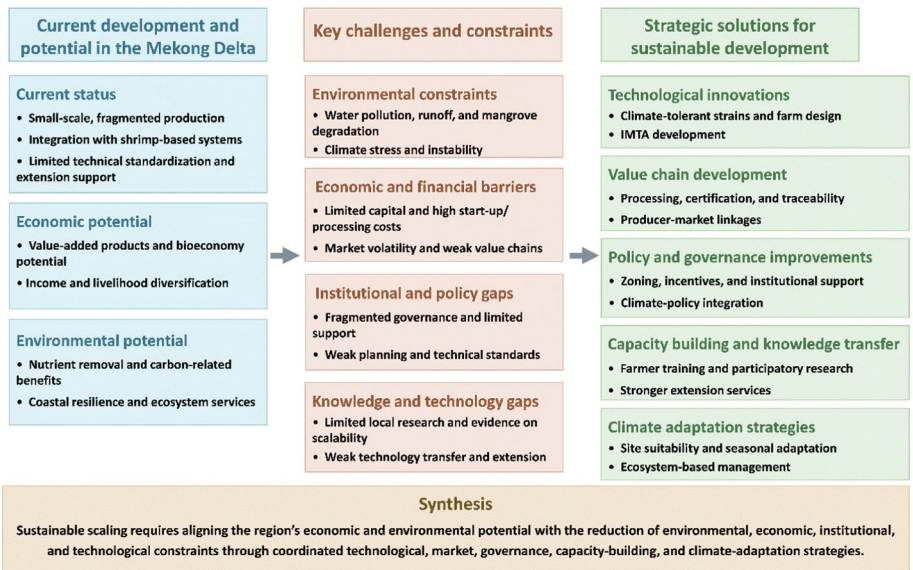


Fig. 3 Schematic synthesis corresponding to Sects. 4., 5. and 6.: current development and potential, key challenges and constraints, and strategic solutions for sustainable seaweed aquaculture development in the Mekong Delta. Source: Authors' synthesis based on the reviewed literature

Table 1 Synthesis of development opportunities, key constraints, strategic responses, and policy implications for sustainable seaweed aquaculture in the Mekong Delta

Dimension	Key opportunities	Major constraints	Strategic responses	Policy implications
Economic	- Value addition through hydrocolloids, functional foods, pharmaceuticals, and bio-based materials - Income diversification for coastal households - Potential contribution to a coastal bioeconomy	- Limited access to capital and credit - High start-up and processing costs - Market volatility and weak value-chain coordination	- Invest in processing technologies and post-harvest infrastructure - Improve certification, traceability, and quality assurance systems - Strengthen linkages among producers, processors, traders, and export markets	Promote seaweed aquaculture as a strategic coastal livelihood and value-added industry rather than a low-value supplementary activity
Environmental	- Nutrient removal from aquaculture effluents and agricultural runoff - Climate-related benefits through carbon uptake - Support for coastal resilience and ecosystem-service provision	- Water pollution and chemical runoff - Mangrove degradation associated with coastal aquaculture expansion - Climate-related stressors affecting yield, biomass quality, and cultivation stability	- Develop and adopt climate-resilient strains and improved farm design - Expand IMTA and ecosystem-based management approaches - Integrate site suitability assessment and water-quality planning into cultivation strategies	Position seaweed aquaculture within coastal environmental management, marine spatial planning, and nature-based climate adaptation strategies
Social	- Livelihood diversification for coastal communities - Greater resilience for climate-vulnerable households - Potential for gender inclusion and local participation across the value chain	- Limited technical skills and practical knowledge among producers - Weak extension, training, and advisory support - Uncertain adoption at the community level	- Expand farmer training, extension services, and technical support - Promote participatory research and co-learning among stakeholders - Strengthen local implementation capacity and community engagement	Embed seaweed aquaculture in inclusive rural and coastal development programs with strong community participation and capacity building
Institutional/governance	- Better integration into coastal and marine planning - Opportunity to align with blue economy and climate adaptation policies - Potential to improve coordination across coastal development sectors	- Fragmented governance and overlapping institutional responsibilities - Limited sector-specific policy support - Weak spatial planning, technical standards, and regulatory guidance	- Improve aquaculture zoning and coastal spatial planning - Provide targeted incentives, technical assistance, and institutional support - Align seaweed aquaculture policy with climate adaptation and coastal governance frameworks	Create an enabling governance environment that reduces uncertainty, strengthens coordination, and supports long-term sectoral scaling

IMTA integrated multi-trophic aquaculture. Source: Authors' synthesis based on the reviewed literature

implications and priority research needs for supporting the sustainable scaling of seaweed aquaculture in the Mekong Delta.

7 Policy implications and research outlook

7.1 Policy implications

The findings of this review suggest that seaweed aquaculture should be recognized as a strategic sector within the green and blue economy, particularly in climate-vulnerable coastal regions such as the Mekong Delta [14, 55]. Given its multifunctional capacity to generate economic benefits, deliver ecosystem services, and support climate adaptation, seaweed farming can contribute meaningfully to more sustainable coastal development pathways [14]. To realize this potential, seaweed aquaculture should be more systematically integrated into coastal management frameworks, including aquaculture zoning, marine resource governance, and coordination with mangrove conservation and restoration efforts [43]. Such integration would help reduce resource-use conflicts, improve environmental outcomes, and support longer-term sector development. In addition, seaweed aquaculture should be incorporated more explicitly into national and regional climate adaptation strategies, particularly in recognition of its role as a nature-based solution [50].

Policy frameworks should therefore move beyond general support and instead provide targeted instruments for sector development. These may include financial incentives, technical assistance, and institutional mechanisms that facilitate public–private partnerships, producer participation, and community-based implementation [50]. Within the Mekong Delta, a coherent and forward-looking policy framework is especially important, as the long-term viability of seaweed aquaculture depends not only on biological feasibility, but also on the alignment of governance, investment, and coastal adaptation planning. In this sense, seaweed aquaculture should be viewed not as a marginal supplementary activity, but as a potentially strategic component of climate-resilient coastal transformation.

7.2 Limitations of the current evidence base and future research outlook

The current evidence base on seaweed aquaculture in the Mekong Delta remains limited, fragmented, and uneven across environmental, economic, social, and institutional dimensions. Much of the available knowledge on ecosystem services, market development, and cultivation technologies has been derived from studies conducted in other coastal regions. Although these findings provide useful comparative insights, their direct applicability to the ecological conditions, production systems, market structures, and governance context of the Mekong Delta has not yet been sufficiently validated. In addition, the lack of long-term field studies, standardized assessment methods, and locally grounded economic data limits the extent to which the productivity, environmental performance, and scalability of seaweed aquaculture can currently be assessed with confidence.

Against this background, several research priorities should be addressed to support evidence-based policymaking and sustainable sector development. First, more robust quantification of the ecosystem services provided by seaweed farming is needed, particularly with regard to carbon-related climate benefits, nutrient removal, and coastal

protection [56]. Such evidence is essential for determining whether, and under what conditions, these benefits can be incorporated into environmental accounting, policy evaluation, climate-finance mechanisms, and investment decisions. Further research should also distinguish short-term carbon uptake by cultivated biomass from long-term carbon sequestration to avoid overstating the potential climate-mitigation contribution of seaweed farming.

Second, improved economic valuation approaches are required to capture both the market and non-market benefits of seaweed aquaculture. A more comprehensive valuation framework could clarify the sector's broader contribution to sustainable development while providing a stronger basis for public investment, financial incentives, and policy integration [57]. Future assessments should consider not only production revenues, but also establishment and operating costs, market volatility, processing requirements, ecosystem-service benefits, livelihood effects, and the distribution of costs and benefits among farmers, processors, communities, and other stakeholders.

Third, more research is needed on the long-term environmental effects of seaweed cultivation. Although seaweed farming may provide nutrient-removal and habitat-related benefits, large-scale expansion could also generate unintended ecological consequences, including changes in nutrient cycling, light availability, hydrodynamic conditions, habitat structure, and interactions with surrounding species. These potential effects should be examined through longitudinal, site-specific, and system-based assessments before extensive cultivation is promoted [44]. Particular attention should also be given to carrying capacity, site suitability, species selection, disease risks, and the cumulative impacts of seaweed farms when combined with other coastal production activities.

Finally, the conditions under which IMTA systems can be adapted and scaled up in the Mekong Delta require closer examination. In these systems, seaweed is co-cultivated with species such as shrimp or fish to recycle dissolved nutrients and improve resource-use efficiency. However, their wider adoption depends on locally demonstrated technical feasibility, economic viability, environmental performance, farmer acceptance, and institutional support [17]. Future studies should therefore evaluate appropriate species combinations, farm configurations, stocking and harvesting practices, biosecurity measures, and governance arrangements under the salinity, temperature, hydrological, and production conditions of the Mekong Delta.

The future development of seaweed aquaculture in the region will consequently depend on the generation of a stronger, locally validated, and interdisciplinary evidence base. Priority should be given to long-term field experiments, pilot-scale production models, comparative economic assessments, carrying-capacity evaluations, and participatory studies involving farmers, researchers, industry actors, and policymakers. Such research would help determine where seaweed aquaculture is environmentally appropriate, economically viable, and socially acceptable. In parallel, technological innovation should be aligned with market development and institutional support so that promising cultivation models can progress beyond isolated or small-scale initiatives. Rather than assuming that seaweed aquaculture is universally suitable across the Mekong Delta, future research and policy should identify the specific locations, production systems, value chains, and governance conditions under which it can contribute effectively to climate-resilient and inclusive coastal development.

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Data availability
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Declarations

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